

Math 421

Friday, November 14

Announcements

- Makeup exam this evening –
please avoid discussing exam

Chapter 3: Continuity

Section 18: Properties of Continuous Functions

Bounded Functions

Definition: A real-valued function f is said to be **bounded** if

- ❖ Give an example of a function that is bounded on $\mathbb{R}$ and a function that is unbounded on $\mathbb{R}$.

Discussion Questions

- ❖ What would it mean for a real-valued function to be *unbounded*?
- ❖ Give an example of a bounded function.
- ❖ Give an example of an unbounded function.
- ❖ Give an example of a function that is bounded below, but not bounded above.

Extreme Value Theorem

Theorem 18.1. Let f be a continuous real-valued function on a closed interval $[a, b]$. Then

- f is a bounded function.
- f assumes its maximum and minimum values on $[a, b]$.

Discussion Questions

- ❖ Remove the continuity restriction on f from the hypotheses of the theorem. Does the conclusion still hold? If not, find (or sketch) a counterexample.
- ❖ Replace the closed interval, $[a, b]$, with the open interval, (a, b) . Does the conclusion still hold? If not, find (or sketch) a counterexample.

Proof Idea

Boundedness:

Proof Idea

Maximum Attained:

Intermediate Value Theorem

Thm. If f is a continuous real-valued function on an interval I , then f has the intermediate value property on I :

Proof Idea

Example

Let $f: [0,1] \rightarrow [0,1]$ be a continuous function. Show that there is some $x_0 \in [0,1]$ such that $f(x_0) = 1 - x_0$.